

MALMO, Sweden

WMO: 026350

Lat: 55.5715N

Lon: 13.0708E

Elev: 69

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	19.9	10.5	9.4	18.0	14.7	11.6	22.0	25.1	40.6	22.5	39.9	3.5	60	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	79.9	65.9	77.0	65.2	73.9	63.5	68.5	75.9	66.7	73.4	65.0	71.4	8.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	95.1	71.6	64.1	89.8	69.9	62.5	84.8	68.5	32.8	76.1	31.3	73.8	30.1	71.4	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.1	15.2	DB	8.7	84.8	7.1	2.9	3.6	86.9	-0.5	88.6	-4.5	90.3	-9.7	92.4	(4)
(5)			WB	8.4	72.2	6.9	2.2	3.4	73.8	-0.7	75.1	-4.6	76.3	-9.6	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	33.6	33.8	37.5	46.0	53.9	59.9	64.0	63.5	57.6	49.2	42.3	36.8		
	DBStd	12.38	6.99	6.41	6.38	5.78	5.51	4.64	4.21	4.14	4.55	5.82	5.88	6.94		
	HDD50	2251	510	454	387	147	21	1	0	0	3	83	234	411		
	HDD65	6200	975	874	852	572	345	164	70	78	225	488	681	876		
	CDD50	1616	0	0	0	25	144	298	435	420	230	60	4	0		
	CDD65	88	0	0	0	0	2	11	40	33	2	0	0	0		
	CDH74	621	0	0	0	1	22	107	318	167	6	0	0	0		
	CDH80	73	0	0	0	0	1	12	43	17	0	0	0	0		
(14) Wind	WSAvg	7.2	8.3	7.9	7.9	7.7	7.1	6.9	6.4	6.3	6.6	6.6	7.4	7.7		
(15) Precipitation	PrecAvg	26.9	2.0	1.5	1.5	1.5	1.9	2.5	2.6	3.1	2.5	2.5	2.7	2.5		
	PrecMax	35.5	3.9	3.5	3.1	2.6	4.5	5.2	8.2	8.7	6.1	4.9	6.1	5.3		
	PrecMin	19.8	0.0	0.3	0.4	0.5	0.6	0.0	0.1	0.5	0.9	0.8	0.6	1.1		
	PrecStd	3.9	1.1	0.6	0.8	0.6	0.8	1.1	1.8	2.2	1.4	1.0	1.2	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	48.3	56.7	69.2	76.9	81.4	84.3	82.1	74.6	64.0	55.7	50.2		
		MCWB	46.5	45.0	48.4	56.8	63.7	65.0	68.4	67.2	64.1	58.0	53.7	48.6		
	2%	DB	45.7	45.8	52.0	64.4	72.7	77.3	80.7	78.4	70.9	60.6	53.3	48.5		
		MCWB	43.9	43.4	45.4	53.5	60.2	64.3	66.5	65.8	62.1	56.8	51.5	46.9		
	5%	DB	44.1	44.0	48.5	60.2	69.1	73.0	77.8	75.2	67.7	58.7	51.3	46.8		
		MCWB	42.5	42.0	44.2	50.6	58.0	61.8	65.6	64.6	60.8	55.8	49.5	45.3		
	10%	DB	42.5	42.3	45.9	56.6	65.3	69.7	74.5	72.1	65.2	56.9	49.7	45.2		
		MCWB	41.0	40.5	42.9	49.0	55.9	60.2	64.2	63.8	59.2	54.2	48.0	43.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	46.0	49.4	58.5	65.7	69.1	71.4	70.8	66.3	59.7	54.2	49.4		
		MCDB	47.8	47.3	53.8	67.6	73.9	78.4	81.3	78.1	71.9	61.9	54.9	49.9		
	2%	WB	44.5	43.9	46.6	54.3	62.1	65.4	68.9	68.1	63.4	57.5	51.3	47.3		
		MCDB	45.2	45.3	49.9	62.4	70.3	73.2	76.8	74.5	69.0	59.8	52.4	48.2		
	5%	WB	42.7	42.3	44.9	51.8	59.4	63.2	67.2	66.2	61.6	56.1	49.6	45.6		
		MCDB	43.8	43.8	47.8	58.5	66.3	70.6	74.6	72.1	66.1	58.2	50.9	46.7		
	10%	WB	41.0	40.6	43.2	49.5	56.9	61.3	65.4	64.7	60.1	54.7	48.1	44.0		
		MCDB	42.4	42.1	45.9	55.1	63.7	68.2	72.2	70.2	64.1	56.6	49.3	45.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.4	10.9	15.3	16.4	15.0	14.9	14.2	12.7	10.0	7.1	6.6		
		MCDBR	6.3	8.0	14.7	22.0	23.0	21.8	22.8	19.4	16.4	11.1	7.0	6.6		
	5% WB	MCWBR	6.2	6.2	9.8	13.0	12.8	11.2	10.8	9.3	9.3	8.0	6.3	6.5		
		MCWBR	6.3	7.6	13.3	18.8	19.5	18.6	18.9	16.1	15.0	10.0	6.7	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.338	0.360	0.393	0.390	0.384	0.405	0.407	0.375	0.358	0.339	0.306		
	taud		2.285	2.304	2.297	2.271	2.340	2.381	2.341	2.346	2.416	2.395	2.331	2.258		
	Ebn at Noon		188	230	254	262	270	273	264	255	248	220	178	161		
	Edh at Noon		17	25	31	37	36	36	36	34	28	23	17	14		
(44) All-Sky Solar Radiation	RadAvg		142	335	772	1329	1710	1820	1690	1360	963	473	181	95		
	RadStd		27	53	86	161	171	140	206	100	94	69	32	9		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-302	+112	N/A

Nomenclature: See separate page